



BlockChain Tools

Smart Contract Audit



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PREPARED FOR

Blockfi Ai

Introduction

Auditing Firm	BlockchainTools
Client Firm	BlockFi
Methodology	Automated Analysis, Manual Code Review
Contract	0xd423c756ca7967a34874cb537e1039a8a2bf4a281127dc12b71f9c718065d48b::bfi::BFI
Blockchain	SUI Blockchain
Mutability	Immutable
Package ID	0xd423c756ca7967a34874cb537e1039a8a2bf4a281127dc12b71f9c718065d48b
Website	https://blockfi-ai.com
Telegram	https://t.me/Blockfi_ai
X (Twitter)	https://x.com/blockafi
Gitbook	https://github.com/blockafi
Report Date	31 March 2025

Verify the authenticity of this report on our website: <https://www.github.com/>



EXECUTIVE SUMMARY

BlockchainTools has conducted both automated and manual analyses of the source code, reviewing it for common contract vulnerabilities and potential centralized exploits. Here's a brief audit summary:

Status	Critical	Major	Medium	Minor	Unknown
Open	0	0	0	0	0
Acknowledged	0	0	0	0	0
Resolved	0	0	1	0	0

- Please be aware that smart contracts deployed on blockchains are not immune to exploits, vulnerabilities, or hacks. Blockchain and cryptographic assets rely on emerging technologies, which inherently carry significant risks. For a comprehensive understanding of risk severity, source code vulnerabilities, and audit limitations, we recommend reviewing the audit report thoroughly.
- Additionally, centralization privileges—regardless of their inherent risk level—can significantly impact the safety and security of smart contracts.



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SCOPE OF WORK

- BlockchainTools was consulted by BlockFi to conduct the smart contract audit of their source codes. The audit scope of work is strictly limited to mentioned token only:
- 0xd423c756ca7967a34874cb537e1039a8a2bf4a281127dc12b71f9c718065d48b::bfi::BFI
- If source codes are not deployed on the main net, they can be modified or altered before main-net deployment. Verify the contract's deployment status below:
- Token Link

<https://suiscan.xyz/mainnet/coin/0xd423c756ca7967a34874cb537e1039a8a2bf4a281127dc12b71f9c718065d48b::bfi::BFI/holders>

Token Name	BlockFI-AI
Owner Program	0x5d8404655eb518747057b05eafd97b6a6f908699049a130cbc842a5d251765c3
Current Supply	1B Token
Decimals	9
Token Extensions	False



AUDIT METHODOLOGY

Smart Contract Auditing Process on Sui Blockchain

Smart contract audits on the Sui blockchain follow a structured methodology to ensure security, efficiency, and compliance with best practices. Mutual collaboration is essential for an effective audit. Below is an overview of BlockFi's auditing process and methodology for Sui-based smart contracts:

CONNECT

- The onboarding team gathers Move language source code and contract specifications to understand the scope and complexity of the audit.

AUDIT

- *Automated Analysis:*
 - Performed using specialized tools to detect vulnerabilities in Move-based smart contracts.
 - Common tools and frameworks include:
 - Sui Move Analyzer (for contract structure and security checks)
 - SWC Vulnerabilities Registry (for identifying known smart contract weaknesses)
 - Formal Verification Tools (to ensure secure execution and prevent logical errors)
- **Simulation & Stress Testing:**
 - Transaction simulations are executed to detect centralization risks, reentrancy attacks, and state inconsistencies.
 - Testing is conducted to identify potential execution failures and trade lock risks on Sui's object-based model.
- **Manual Code Review:**
 - A line-by-line inspection is performed to detect vulnerabilities related to object ownership, access control, and transaction dependencies.
 - The audit specifically reviews risks associated with:
 - Move module permissions (ensuring no unauthorized access)
 - Object mutability risks (to prevent unintended changes)
 - Storage and gas optimization (to enhance efficiency and cost-effectiveness)



REPORT

The auditing team prepares a preliminary report detailing all performed checks and identified findings.

The client's development team reviews the report and implements necessary amendments to the Move source code.

After revisions, the auditing team delivers a final comprehensive report, highlighting any open or unresolved issues.

PUBLISH

The client can choose to use the audit report internally or make it publicly available.

Note: Smart contract audits on the Sui blockchain do not have a pass or fail status. Instead, they serve as an objective assessment of the security and robustness of the source code.

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RISK CATEGORIES

Smart contracts are primarily designed to hold, approve, and transfer tokens, making them attractive targets for attackers.

External Attacks: A successful attack from an external entity may result in direct exploitation of the contract.

Centralization Exploits: If centralization risks exist, a privileged role may have the ability to manipulate the contract for personal gain.

All identified risks in the audit report are categorized below for reference and further review.

Risk Type	Definition
Critical	These risks could be exploited easily and can lead to asset loss, data loss, asset, or data manipulation. They should be fixed right away
Major	These risks are hard to exploit but very important to fix, they carry an elevated risk of smart contract manipulation, which can lead to high-risk severity.
Medium	These risks should be fixed, as they carry an inherent risk of future exploits, and hacks which may or may not impact the smart contract execution. Low-risk re-entrancy-related vulnerabilities should be fixed to deter exploits.
Minor	These risks do not pose a considerable risk to the contract or those who interact with it. They are code-style violations and deviations from standard practices. They should be highlighted and fixed nonetheless.
Unknown	These risks pose uncertain severity to the contract or those who interact with it. They should be fixed immediately to mitigate the risk uncertainty.



METADATA ANALYSIS

Metadata analysis on the Sui blockchain involves examining the Move-based metadata structure of assets. Key attributes include object ownership, creator details, access controls, and linked multimedia files. This process ensures authenticity, integrity, and compliance with Sui's decentralized standards. It helps detect unauthorized modifications or security risks in asset management. By verifying metadata, we enhance transparency and trust in Sui-based digital assets.

Key Attributes Analyzed

Object Ownership – Tracks the ownership of assets.

Creator Details – Verifies the creator's identity.

Access Controls – Ensures appropriate permissions for asset modification and transactions.

Multimedia Files – Validates linked digital content (images, videos).

Minting & Transaction Details – Confirms compliance with minting and transaction rules.

Conclusion

The analysis ensures that assets on the Sui blockchain are secure, comply with decentralization standards, and maintain transparency, enhancing trust and integrity within the ecosystem.



DEFINITION

Record Type	Identifies the type of record within the Sui blockchain structure (e.g., asset, token, etc.).
UpdateAuthority	Account authorized to update the asset's metadata.
Mint	Represents the account responsible for creating this token or asset.
Data	Contains specific details and attributes about the asset or token.
Name	The name associated with this asset or token.
Symbol	Trading symbol of this asset or token.
URI	A link to external information related to the asset, typically hosted on IPFS or another decentralized storage solution
SellerFeeBasisPoints	The sales fee for this token or asset, measured in basis points.
PrimarySaleHappened	Indicates whether this asset or token has been sold for the first time.
IsMutable	Specifies if metadata can be modified after creation. Once set to False, metadata cannot be changed again.
EditionNonce	An optional field used to distinguish different editions of the token or asset.
TokenStandard	Defines the compliance standard of the token
Description	A textual description of the token or asset, often providing additional context or details.
Image	A link to an image or visual representation of the asset



KEY INFORMATION

CreateAccount transaction

<https://suiscan.xyz/mainnet/tx/EJBXoYwEbfjHNzQyRtXj4oCrQJkkdhBdHgWgJrpaoYaq>

totalSupply

1 Billion

New mintTokens authority

NULL

New freezeAccount authority

NULL

isMutable

0



MANUAL REVIEW

Identifier	Definition	Severity
CEN-02	Initial token distribution	Medium

Centralized role holds substantial portion of the circulating supply. Large number of tokens in one address raises significant concerns about centralization within the token’s ecosystem.

Balance Change				85 RAW
ID	Type	Change	Currency	
0x5d8404655e...2a5d251765c3	Account	- 2.09039488 4.70338803 USD	SUI	
0x5d8404655e...2a5d251765c3	Account	+ 1,000,000,000	BFI	
@winterwalrus 0xdd224f2287...611896b007e6	Account	+ 2 4.5 USD	SUI	
Object Change				85 RAW
Object ID	Current Owner	Type	Status	Version
0x3ed84d1332e...88eb974c5e777	Sui System account 0x000000...00000000	0x2:...BFI>	Created	510278108
0x42ef8fd5191...6eb101941f4c2	0x5d8404...251765c3	0x2:...BFI>	Created	510278108
0x57a4d179c9e...d43a17e16f02f	@winterwalrus 0xdd224f...96b007e6	0x2:...SUI>	Created	510278108
0x5aaa09290a5...005c80b81808a	shared(510278108)	0x2:...BFI>	Created	510278108
0x9cd2acbee2...fe8f46ed5f370	0x5d8404...251765c3	0x2:...eCap	Created	510278108
0x219e45090f1...7382c86ff1994	0x5d8404...251765c3	0x2:...SUI>	Updated	510278108
0xd423c756ca7...f9c718065d48b	immutable	-	Published	1

RECOMMENDATION

Project must communicate with stakeholders and obtain the community consensus while distributing tokens.

RESOLUTION

BlockFi team will distribute tokens as per their pre-determined tokenomics.



DISCLAIMERS

BlockchainTools provides an easy-to-understand audit of blockchain source codes (commonly known as smart contracts). The smart contract for this particular audit was analyzed for common contract vulnerabilities and centralization exploits. This audit report makes no statements or warranties on the security of the code.

This audit report does not provide any warranty or guarantee regarding the absolute bug-free nature of the smart contract analyzed, nor do they provide any indication of the client's business, business model, or legal compliance. This audit report does not extend to the compiler layer, any other areas beyond the programming language, or other programming aspects that could present security risks.

Cryptographic tokens are emergent technologies, and they carry high levels of technical risks and uncertainty. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis. This audit report could include false positives, false negatives, and other unpredictable results.

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About BlockchainTools

BlockchainTools provides intelligent blockchain solutions. We specialize in smart contract development, testing, and auditing services. With over 150+ solidity codes developed, 1000+ smart contracts audited, and over 500,000+ lines of code analyzed, we have extensive experience working on major public blockchains such as Ethereum, Binance, Cronos, Doge, Polygon, Avalanche, Metis, Fantom, Bitcoin Cash, Velas, Oasis, and more.

BlockchainTools is built by engineers, developers, UI experts, and blockchain enthusiasts. Our team currently consists of 4 core members, along with 6+ casual contributors.

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